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An Agenda for Studying Credibility Perceptions of Visual Misinformation

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ABSTRACT

Today's political misinformation has increasingly been created and consumed in visual formats, such as photographs, memes, and videos. Despite the ubiquity of visual media and the growing scholarly attention to misinformation, there is a relative dearth of research on visual misinformation. It remains unclear which specific visual formats (e.g., memes, visualizations) and features (e.g., color, human faces) contribute to visual misinformation's influence, either on their own or in combination with non-visual features and heuristics, and through what mechanisms. In response to these gaps, we identify a theoretical framework that explains the persuasive mechanisms and pathways of visual features in lending credibility (e.g., as arguments, heuristics, and attention determinants). We propose a list of relevant visual attributes to credibility perceptions and a research agenda that integrates methods including computational visual analysis, crowdsourced annotations, and experiments to advance our understanding of visual misinformation.

KEYWORDS

Visual misinformation; misinformation; credibility; image; video

The functioning of a healthy democracy critically depends on its citizens being factually informed. The increasing proliferation of misinformation in today's media environment threatens national security, social cohesion, and public health. For example, misinformation about election results may threaten the democratic process and cause social unrest. Misinformation about coronavirus disease 2019 (COVID-19) and vaccination has resulted in otherwise preventable public health crises and loss of lives around the world. In particular, today's misinformation posts have increasingly been created and consumed in multimodal formats, including photos, memes, videos, and data visualizations. For example, Li et al. (2022) found that 11% of YouTube's most popular videos on COVID-19 vaccines contain misinformation, and Garimella and Eckles (2020) found that visual misinformation was prevalent among images circulated on WhatsApp in India. The superior impression, retention, virality, and persuasiveness of visual information all warrant scrutiny from misinformation researchers (e.g., Keib et al., 2018; Y. Li & Xie, 2020; Peng, 2018; Powell et al., 2015).

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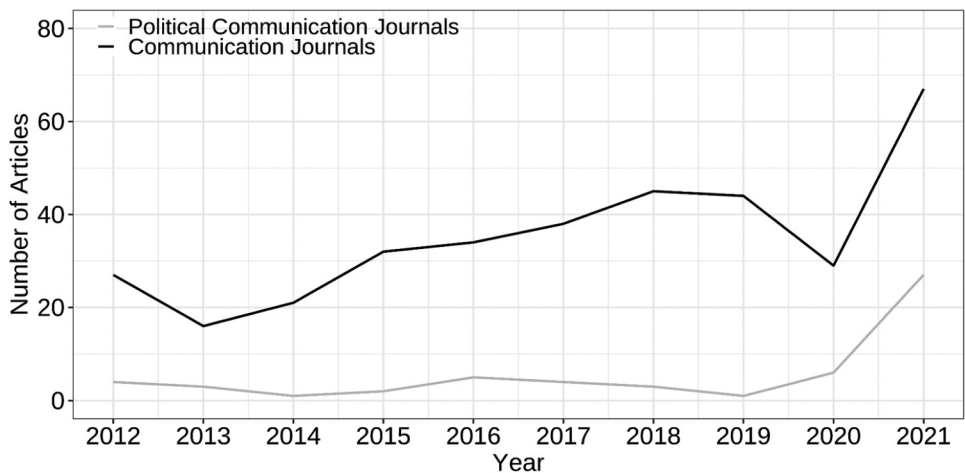


Figure 1. Visual-related scholarship in flagship communication journals.
Notes: We identified articles from three flagship political communication journals (*Political Communication*, *The International Journal of Press/Politics*, and *Public Opinion Quarterly*) and five leading communication journals (*Journal of Communication*, *Journal of Computer-Mediated Communication*, *New Media & Society*, *Communication Research*, and *Human Communication Research*). We searched for the keywords “image,” “video,” “visual,” and “multimodal” in their titles, keywords, and abstracts for visual-related studies. The publication year was decided by when the article was first published online.

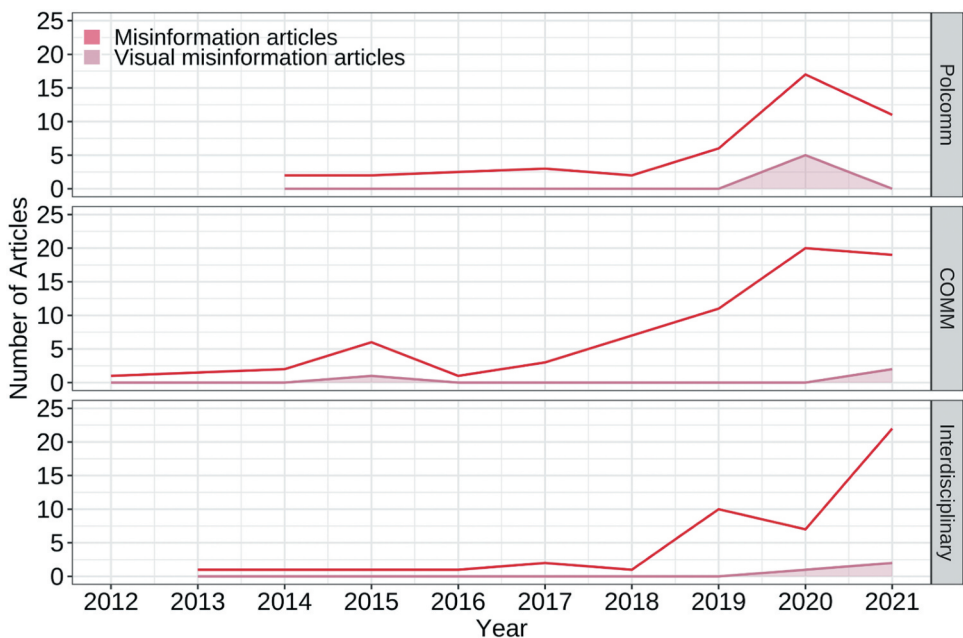


Figure 2. Scholarship on misinformation and visual misinformation in flagships journals in three domains.
Notes: In addition to the flagship political communication journals and leading communication journals we included five interdisciplinary journals (*Nature*, *Science*, *PNAS*, *Nature Human Behavior*, and *Science Advances*). We identified relevant research articles if their title, keywords, or abstract contain “misinfo*,” “disinfo*,” or “fake news.” We included research articles only and excluded opinion articles, book reviews, and editorials. We also searched for the keywords “image,” “video,” “visual,” and “multimodal” for articles studying visual misinformation.

Amid the growing attention to misinformation, we argue that there are two critical gaps that call for more scholarly examination. First, there is a dearth of research on *visual* misinformation, despite the outsized role of visual media in the contemporary media environment. A 40-country study shows that 67% of the online population regularly consume news videos (Newman et al., 2020). Visual-oriented social media apps dominate the social media landscape, with 81% of Americans using YouTube and 40% using Instagram (Auxier & Anderson, 2022). Scholarly research on visual communication has also increased in the past decade (Figure 1). In 2021, for example, 12% of the 223 articles published in flagship journals of political communication contain at least one visual-related keyword.

However, against this backdrop, research on visual misinformation remains disproportionately scant, as most existing studies still focus on *textual* misinformation. In Figure 2, we summarize the number of mis/disinformation-related research articles in flagship political communication journals, leading communication journals, and interdisciplinary journals in the past decade in three separate panels. While all three domains witness a consistent growth of misinformation scholarship, the proportion of visual misinformation remains low (Figure 2). For example, none of the 11 articles on misinformation published in political communication flagship journals in 2021 focused on visual misinformation.

Second, there is a dearth of research on how (1) specific visual formats, such as photographs, memes, and videos, and (2) specific visual features, such as color, facial expressions, and esthetic appeal, may influence the audience's credibility judgments. Content analyses of visual misinformation reveal that a broad range of visual formats and features exist, which may provide different functions in lending credibility to false claims (Brennen et al., 2021; Garimella & Eckles, 2020). Disinformation actors often adopt a wide range of visuals to cater to distinct audience groups (e.g., Bastos et al., 2021). Yet, while there are experimental studies of visual misinformation, they often treat visuals as a black box and compare them against other modalities, such as text (Hameleers et al., 2020; Sundar et al., 2021), or compare between original and manipulated visuals, such as deepfake videos (Dobber et al., 2021; Vaccari & Chadwick, 2020). Therefore, we need to unpack the visual modality and examine how specific types and variations of visual misinformation affect audience perceptions.

In this article, we propose a research agenda to address these two critical gaps in our understanding of visual misinformation. Researchers need to pay more attention to the diffusion and effects of misinformation claims in visual and multimodal formats. Further, we need to understand how specific visual formats and features influence credibility perceptions of such claims. In the following, we propose a theoretical framework that explains the persuasive mechanisms of visual features, provide an overview of relevant visual features, and suggest directions for future research.

Persuasive Mechanisms of Visual Features: A Theoretical Framework

Several theories could be relevant to the persuasive power of visual misinformation. In prior research, indexicality is frequently cited to explain why visual misinformation likely looks more credible than its textual counterparts (e.g., Hameleers et al., 2020). Photographs and videos can be seen as indexical signs of the objects and events portrayed in them and imply an evidential connection between reality and representations (Messaris, 1997). Still,

indexicality hardly applies to noticeably manipulated visuals such as memes and infographics. Scholars have also developed models to explain how users evaluate credibility in online systems (e.g., Metzger & Flanagin, 2015; Sundar, 2008). For example, the Modality-Agency-Interactivity-Navigability (MAIN) model proposes that users' judgments of the credibility of online information are affected by cues related to modality, agency, interactivity, and navigability (Sundar, 2008). When it comes to modality, as the presentation mode enriches realism heuristics from text to audio to video, the messages more closely resemble reality and are seen as more credible (Sundar, 2008; Sundar et al., 2021).

Here, we choose the Elaboration Likelihood Model (ELM) to illustrate the roles that visual features could play in influencing people's credibility judgment. While ELM is not the only relevant theory, it is particularly useful as an overarching framework to organize a broad range of visual formats and attributes. As a dual process model of persuasion, ELM posits that there are multiple routes through which a message can lead to attitudinal changes. Message recipients with both motivation and ability (high elaboration likelihood) are more likely to engage in effortful thinking – the central route defined by ELM – and to base their attitudes on scrutiny of issue-relevant arguments. When the recipients lack the motivation or ability to process information (low elaboration likelihood), they tend to engage in a route that requires less effort – a peripheral route defined by ELM – using peripheral cues or heuristics to form judgments about the issue (Petty & Cacioppo, 1986). ELM has been applied to the credibility context in Metzger's (2007) dual processing model of credibility evaluation, which suggests that user motivation and ability may determine the mode of credibility evaluation. In other words, the more motivated and capable users are, the more likely they will be to evaluate credibility systematically, rather than heuristically.

In ELM, a message variable can serve three potential roles: (1) as an argument that individuals under high elaboration can scrutinize to form their judgment, (2) as a heuristic that individuals under low elaboration can rely on, and (3) as an attention determinant that shapes the amount of effortful thinking that individuals spend on a certain message (Petty & Cacioppo, 1986). Similarly, we argue that the visual features of misinformation posts might also influence people's credibility perception in these three pathways:

- (1) **Visual features as arguments.** Examples such as data visualizations and text overlays might contain actual arguments that people would effortfully consider in order to form impressions.
- (2) **Visual features as heuristics.** Visual features such as high esthetic appeal might act as a heuristic that suggests the message is from a professional and credible source, which leads people to give high credibility perceptions. Visuals presented as photographs and videos might signal a realism heuristic and make people more likely to believe it is true (Sundar et al., 2021).
- (3) **Visual features as attention determinants.** Visual elements such as arousing color might make a message more likely to get viewers' attention and provoke emotional arousal (Morgan et al., 2003), thus leading them to closely engage with the message content. Also, visual-based format such as videos might be too distracting for viewers, which could prevent them from systematically scrutinizing content (Sundar et al., 2021).

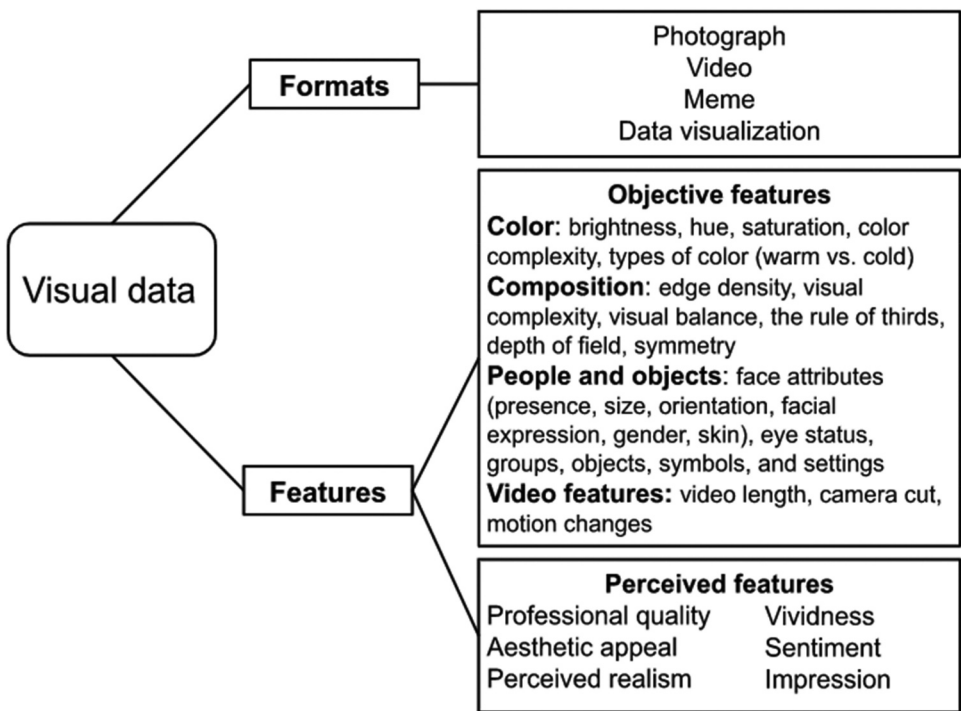


Figure 3. A summary of visual formats and features.

An Overview of Visual Formats and Features

Next, we discuss different **visual formats** most commonly seen in the online (mis)information context (Figure 3), followed by a distinction between **objective** and **perceived** visual features, informed by prior research (Kim, 2015; O’Keefe, 2003; Tao & Bucy, 2007). Objective features are intrinsic properties of visual messages and exist independently of audience perceptions, such as brightness and the presence of faces. Perceived features (or effect-based features) reflect mediating psychological states and summarize the audience’s reactions to visual messages, which can be calculated by aggregating viewers’ subjective evaluations of a message (e.g., esthetic appeal; Kim, 2015).

Visual Formats

We highlight four common visuals that we mostly encounter online: (1) photographic images, (2) visualizations, (3) image memes, and (4) videos. **Photographic images** portray reality through still images. As photographic images contain a high level of realism heuristics, they could improve the credibility of and engagement with the information (Hameleers et al., 2020; Y. Li & Xie, 2020; Sundar, 2008). **Visualizations**, such as narrative visualizations, cartoons, and infographics, are widely used in journalism (Segel & Heer, 2010), science communication (N. Li et al., 2018), and elections (Amit-Danhi & Shifman, 2022). Research on user interfaces and data visualization shows that visual displays and formatting strategies also affect perceived credibility (Henke et al., 2020), have greater

persuasive power, and elicit less defensive reactions and higher favorability toward content (Oh et al., 2018; Pandey et al., 2014). **Image memes** are an important type of memes, which are “cultural units that spread from person to person” (Shifman, 2013). Because of the derivative and viral nature of memes, especially on the Internet, users may have different expectations and experiences processing memes than they do with other visual formats, making them a uniquely popular conduit for misinformation. **Videos**, with more modalities (visual, audio, and text) compared with other visual formats, cue more heuristics and create additional affordances that evoke greater attention and trust (Sundar & Limperos, 2013). In the misinformation context, Sundar et al. (2021) found that doctored video news segments are more readily believed and shared on social media than text or audio formats. Young et al. (2018) find that fact-checking videos are more effective than long-form fact-checking articles in correcting beliefs and reducing misperceptions.

Objective Features: Color, Composition, and Content

Following prior literature in communication, computer science, and psychology (e.g., Lu & Pan, 2022; Machajdik & Hanbury, 2010; Matz et al., 2019), we highlight three types of objective visual features: (1) color, (2) composition, and (3) content. Among all objective features, **color** features have been widely explored using various indicators. Simple values like brightness and saturation can be directly extracted from the colorspace, and these features were found to affect attention (Camgöz et al., 2004), visual working memory (Qian et al., 2018), and the viewer’s sense of pleasure (Lichtlé, 2007). Different colors may be associated with various human affect and behaviors, with warm colors associated with more positive feelings than cool colors (Bakhshi et al., 2015; Peng & Jemmott, 2018). Research in marketing science also shows that color features may affect engagement with advertisements, consumer choice, and buying intentions (Pelet & Papadopoulou, 2012; Sokolik et al., 2014).

Composition captures how visual elements are organized in an image, such as *edge density*, *visual balance*, *the rule of thirds*, *depth of field*, and *symmetry* (Machajdik & Hanbury, 2010; Matz et al., 2019). Previous research has shown that composition features can predict images’ esthetic appeal (Ke et al., 2006; C. Li & Chen, 2009; Savakis et al., 2000), sentiment (Machajdik & Hanbury, 2010), and diffusion on social media (Totti et al., 2014). For example, Matz et al. (2019) find that extroverted people prefer images with lower-density edges and details.

Content features in visual media can shape viewers’ impressions toward them, which may include perceived trustworthiness and credibility. For example, the *presence of human faces* is associated with higher engagement with visual content on social media (Y. Li & Xie, 2020). Specific face-related features such as orientation, size, and eye status also affect viewers’ perceptions toward the person portrayed (Talamas et al., 2016). *Facial expressions* shape viewers’ perception of people and also credibility judgments (Ert & Fleischer, 2020; Krumhuber et al., 2007). Besides single-person features, *groups*, *objects*, *symbols*, and *settings* also influence how we perceive visuals. Empirical studies found a “cheerleader effect” in which people in *group pictures* were perceived as more attractive than individuals photographed alone (Walker & Vul, 2014). Objects and symbols such as national flags also led viewers to score highly on symbolic patriotism and racial prejudice, and studies found

that settings where a person is presented can effectively shape viewers' impressions (Joo et al., 2014) or social media engagement (Peng, 2021).

Perceived Features

Various perceived features could be relevant. For example, **perceived visual quality** assesses whether the visual appears to be produced by amateurs versus professionals. On one hand, a visual that looks like it is produced by a professional source could make it look more credible. On the other hand, in the age of social media, visuals that look amateurish could be seen as more authentic and more credible. Another important feature is **impressions of people portrayed**. As reviewed earlier, the nonverbal behaviors of the depicted figures (e.g., smiling), associated objects (e.g., a syringe) and settings (e.g., hospital rooms) likely shape viewers' impressions of these figures (Grabe & Bucy, 2009; Peng, 2018, 2021). Such impressions could additionally shape how viewers evaluate messages and arguments embedded in visual posts. We note that individuals' impressions of others are often associated with prejudice and bias (e.g., racism), which could be adopted by malicious political actors to appeal to certain demographic groups (Bastos et al., 2021).

Directions for Future Research

As discussed above, the visual formats and features relevant to the misinformation context provide exciting yet untapped opportunities to advance scholarship in misinformation and visual communication. Next, we identify five promising directions for future research.

- (1) **Quantifying the patterns and prevalence of visual formats and features in the misinformation landscape.** We suggest a combination of computational visual analysis and traditional content analysis to study visual features. Advances in computer vision provide researchers with computational tools to extract objective visual features that may be difficult for human eyes to assess, such as the exact values or distributions of colors, facial orientation, and eye gaze (Lu & Pan, 2022; Matz et al., 2019; Peng, 2018; Peng & Jemmott, 2018; Talamas et al., 2016). Computational tools can also be used to quantify more subjective features such as visual esthetics (Peng, 2022) and emotions (Rao et al., 2019). For visual formats and perceived features that are more semantic-based and harder to assess by computer algorithms, researchers can leverage crowdsource ratings to derive less-biased scores for these features. These quantification efforts provide an intuitive understanding of the prevalence of visual strategies, which also sets the stage for further analysis of the effect of these formats and features.
- (2) **Understanding the effects of visual formats and features on credibility perception.** Similar to how prior research measured text-based credibility perception (Luo et al., 2022), we can derive the credibility score of visuals through crowdsource-based human ratings. To investigate the exact visual attributes that influence human judgment of credibility, we can rely on machine-learning-based feature selection methods (e.g., LASSO) to identify a subset of important factors that contribute to the outcome (Matz et al., 2019). Researchers can also use a linear feature summation model to model the hierarchical relationships among attributes (Iigaya et al., 2021),

clustering algorithms to cluster features that are intercorrelated (Huang, 1997; Lu & Shen, 2023), or activation maps to understand important image regions that predict audience perceptions (Joo & Steinert-Threlkeld, 2022).

- (3) **Understanding how visual formats and features interact with nonvisual features (e.g., text, source, virality) and user attributes.** Existing research on misinformation has identified important nonvisual features that may influence credibility perceptions, such as original source and virality metrics (number of likes and shares; Sundar, 2008). User characteristics are also shown to be associated with their susceptibility, such as demographics, digital media literacy, digital imaging experiences, and political affiliation toward the issue depicted in the visual post, among others (Shen et al., 2019). Through factorial experiments, we can observe and test how visual features may interact with other nonvisual features of the misinformation post and how they function differently across different user groups. Such insights will directly inform social media platforms and fact-checking agencies, which could then effectively target their prevention and correction efforts to posts with specific visual elements and to the most vulnerable user groups.
- (4) **Leveraging visual formats and features to design effective correction campaigns to counteract misinformation.** Recent research finds that fact-checking visuals are more effective than long-form fact-checking articles in reducing misperceptions and correcting beliefs (Young et al., 2018). Existing research on corrections has explored factors such as placement (pre-bunking versus debunking; Tay et al., 2021), source (Vraga & Bode, 2017), and type (fact- versus logic-based) of the corrective information (Vraga et al., 2019), but little research has focused on how different visual features may impact the correction process or outcomes (Lu & Shen, 2023). Through experiments, we can begin to identify the most effective combinations of color attributes, facial expressions, and so on for corrections designed for different issue contexts and user groups.
- (5) **Connecting visual credibility to other contexts beyond misinformation.** Finally, we believe knowledge about credibility perceptions of visual misinformation can be fruitfully extended to other contexts and topics in political communication. For example, previous research has focused on visual representations of politicians in partisan media and social media (Grabe & Bucy, 2009; Peng, 2018, 2021; Xi et al., 2020). Visual portrayals of politicians, which are often subject to gender biases (Van der Pas & Aaldering, 2020), could shape voters' impressions of their trustworthiness and competence (Dan & Arendt, 2021; Peng, 2018). Our framework could pinpoint specific elements in such visual portrayals that may sway viewers in certain ways, and suggest concrete steps to implement more fair and equitable visual coverage practices across gender and racial groups. As such, we believe that understanding how visual formats and features may influence viewers' perceptions could significantly advance research on political communication in today's media environment.

Disclosure Statement

All three authors declare that we do not have any relevant financial or non-financial competing interests.

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